

Model Predictive Direct Torque Control for PMSM with Discrete Voltage Vectors

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The direct torque control (DTC) of ac motors leads to a faster torque response with a small number of switching operations in inverters when compared with a conventional approach such as vector control. DTC exhibits a hybrid nature in the sense that the system is composed of continuous variables of torque and flux and involves discrete switching in the inverter. The output of the inverter is limited to the finite discrete values at each instance of sampling. Model predictive control (MPC) is applied to the system so that an optimal switching sequence is derived subject to the given constraints. The proposed MPC-based approach reduces the torque ripple with a small number of switching operations. The effectiveness of the proposed MPDTC approach is verified through simulations and experiments by comparing the proposed approach with conventional DTC.

Keywords: permanent magnet synchronous motor (PMSM), direct torque control (DTC), model predictive control (MPC), optimization

1. Introduction

Several adjustable speed ac motors have been used in various fields with recent advancements in the field of power electronics. In particular, permanent magnet synchronous motors (PMSMs) have received considerable attention, owing to their advantages of low excitation loss, high power factor, high torque per current, and high efficiency. In fact, PMSMs have widely adopted for commercial applications such as electric and hybrid vehicles. Pulse width modulation (PWM) is usually applied in order to control the stator winding current in a PMSM. The switching vector in the inverter is dependent on the PWM carrier frequency. The efficiency of the PMSM drive improves if the carrier frequency is high, but this is not the case owing to the inherent trade-off between control performance and switching loss. To solve the trade-off, direct torque control (DTC) has been proposed as an alternative approach^{(1)–(12)}.

Direct torque control (DTC), which was originally designed

for induction motors (IMs)⁽¹³⁾, yields a faster torque response with a small number of switching operations when compared with conventional vector control⁽¹⁴⁾. DTC for IMs employs hysteresis comparators and a predesignated switching table. The characteristics of DTC are different from those of conventional vector control in that the switching in the inverter is treated as a discrete voltage vector in DTC. The command voltage to the inverter is limited to the finite set of discrete voltages at each instance of sampling, which is always feasible with no modulation such as PWM. From the viewpoint of such a hybrid nature, model predictive control (MPC)^{(15)–(17)} appears to be a promising method for synthesis since an optimal voltage vector can be derived for a given cost function in the presence of constraints on control input and/or state. Model predictive DTC designed for IMs has been proven to reduce both the average number of switching operations and torque ripple^{(18)–(19)}. MPC has also been applied to current control for interior permanent magnet synchronous motors (IPMSMs)^{(17)–(20)–(26)}. Each of the studies has established that the torque response considerably improved in comparison with conventional vector control. Model predictive DTC has also been proposed previously⁽²⁷⁾, which has been shown to effectively reduce switching loss through some simulations.

In this paper, we propose a novel model predictive DTC (MPDTC) approach in order to reduce the torque ripple with a small number of switching operations while achieving a faster response at relatively close speed to a base speed. With the approach, the number of switching operations required for control is explicitly considered, which implies that the trade-off between the number of switching operations and torque ripple can be addressed. In addition, the treatment

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of dead-time in the inverter and rotation of the rotor are also considered for practical use⁽²⁸⁾. The effectiveness of the proposed method is verified through simulations and experiments.

The rest of this paper is organized as follows. Section 2 reviews a mathematical models of the PMSM, vector control, and inverter, followed by an outline of DTC and MPC. In Sect. 3, we propose a novel model predictive DTC approach. In the section, the dead-time and the rotation of the rotor are also discussed for improvement of control performance. The effectiveness of the proposed method is illustrated through simulations and experiments in Sect. 4. Finally, Sect. 5 concludes this paper.

2. Preliminaries

This section reviews the mathematical models PMSM and inverter, focusing on discrete voltage vectors. In addition, DTC and MPC are briefly explained.

2.1 Mathematical Model of PMSM The symbols used in this paper are listed in Table 1.

The PMSM is composed of stator windings arranged in three phases and a permanent magnet rotor. A rotating magnetic field is generated by the three-phase current, and a magnetic torque is generated by the interaction of the rotating magnetic field and the flux of the permanent magnet. In IPMSMs, reluctance torque is also generated owing to magnetic saliency. That is, the IPMSM is driven by the magnetic torque and the reluctance torque.

The d - q coordinate system is composed of two orthogonal axes with rotating coordinates which are synchronized with the rotor. The direction of the field generated by the permanent magnet is along the d -axis, while the q -axis leads the d -axis by 90 [deg] in electrical angle. In contrast, the u - v - w coordinate system is a fixed system composed of u , v and w phases.

The state equation with respect to the d - q current in the PMSM is given as

$$\frac{d}{dt} i_{dq} = \begin{bmatrix} -\frac{R_s}{L_d} & \omega_{re} \frac{L_q}{L_d} \\ -\omega_{re} \frac{L_d}{L_q} & -\frac{R_s}{L_q} \end{bmatrix} i_{dq} + \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix} v_{dq} + \begin{bmatrix} 0 \\ -\frac{\omega_{re} K_e}{L_q} \end{bmatrix}, \dots \quad (1)$$

where the symbols in Eq. (1) are as listed in Table 1. Equation (1) is transformed into its corresponding discrete-time state space equation in an appropriate manner as follows.

$$i_{dq}(k+1) = A i_{dq}(k) + B v_{dq}(k) + f(k), \dots \quad (2)$$

where $k = 0, 1, \dots$

Table 1. Notation

$i_{dq} = \begin{bmatrix} i_d & i_q \end{bmatrix}^T$	stator current vector in d - q axis
$v_{dq} = \begin{bmatrix} v_d & v_q \end{bmatrix}^T$	stator voltage vector in d - q axis
R	stator winding resistance
L_d, L_q	stator winding inductance in d - q axis
K_e	electromotive force constant
P_n	pairs of rotor poles
ω_{rm}	angular velocity of rotor (mechanical angle)
$\omega_{re} = P_n \omega_{rm}$	electric angular velocity of rotor
θ_{re}	rotor pole position (electrical angle)

The torque τ and the magnitude of the magnetic flux ψ are given by

$$\tau = P_n [K_e i_q + (L_d - L_q) i_d i_q], \dots \quad (3)$$

$$\psi = [(K_e + L_d i_d)^2 + (L_q i_q)^2]^{\frac{1}{2}}, \dots \quad (4)$$

respectively, where P_n represents the pairs of rotor poles.

2.2 Three-phase Voltage Source Inverter In vector control⁽¹⁴⁾ for the PMSM, the amplitude and the frequency of the voltage applied to the stator winding are controlled. A three-phase voltage source inverter, as shown in Fig. 1, is generally used as a power supply. The inverter generates an equivalent AC power from a DC source by switching each pair of semiconductor devices such as FETs or IGBTs in the inverter.

In this paper, as shown in Fig. 1, the state when the upper arm is conducted in each phase is denoted by 1 while the opposite is denoted by 0. Note that both the arms are not conducted simultaneously to ensure protection of the devices. The output voltage in each phase with the virtual neutral point is limited to $\frac{E_d}{2}$ or $-\frac{E_d}{2}$, for the dc-link voltage E_d . Then, the instantaneous output voltage of the inverter is confined to the $2^3 = 8$ switching patterns. Each instantaneous inverter switching is shown in Fig. 2. The subscript of the vector in Fig. 2 corresponds to u , v and w phases in sequence. For example, $V_{(000)} = \frac{E_d}{2} \begin{bmatrix} -1 & -1 & -1 \end{bmatrix}^T$, $V_{(100)} = \frac{E_d}{2} \begin{bmatrix} +1 & -1 & -1 \end{bmatrix}^T$, and so forth. Formally,

$$\{V_{(000)}, \dots, V_{(111)}\} = \left\{ \frac{E_d}{2} s : s \in \{-1, 1\}^3 \right\}, \dots \quad (5)$$

In Eq. (2), v_{dq} is given as $v_{dq} = C_{uvw}^{dq}(\theta_{re}) v_{uvw}$ for $v_{uvw} \in \{V_{(000)}, \dots, V_{(111)}\}$. The transformation matrix $C_{uvw}^{dq}(\theta_{re})$ for transforming u - v - w to d - q is defined as follows.

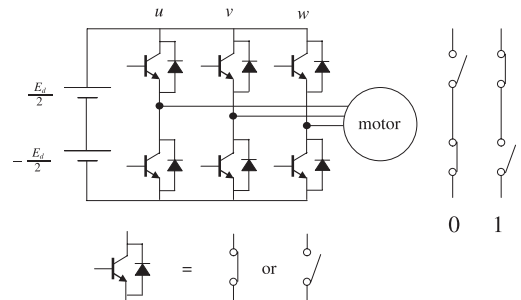


Fig. 1. Three-phase voltage source inverter

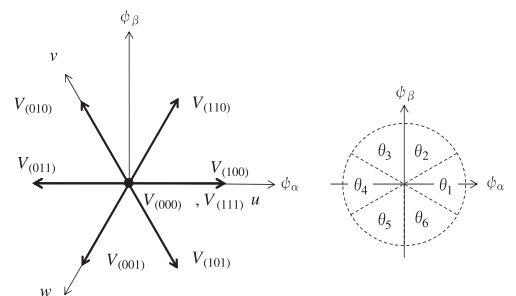


Fig. 2. Space vector and flux position sections

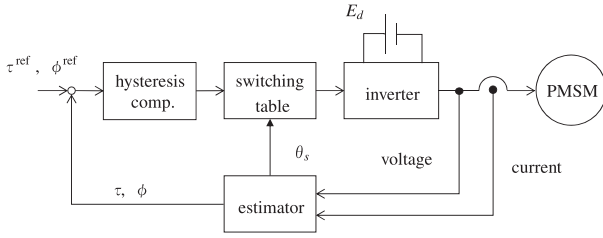


Fig. 3. Block diagram of DTC for PMSM

$$C_{uw}^{dq}(\theta_{re}) = \frac{1}{\sqrt{3}} \begin{bmatrix} \cos \theta_{re} & \cos(\theta_{re} - \frac{2}{3}\pi) & \cos(\theta_{re} + \frac{2}{3}\pi) \\ -\sin \theta_{re} & -\sin(\theta_{re} - \frac{2}{3}\pi) & -\sin(\theta_{re} + \frac{2}{3}\pi) \end{bmatrix} \quad (6)$$

Similarly, $v_{dq} = C_{\alpha\beta}^{dq}(\theta_{re})v_{\alpha\beta}$. The transformation matrix $C_{\alpha\beta}^{dq}(\theta_{re})$ for transforming the static α - β to the d - q system is defined as follows.

$$C_{\alpha\beta}^{dq}(\theta_{re}) = \begin{bmatrix} \cos \theta_{re} & \sin \theta_{re} \\ -\sin \theta_{re} & \cos \theta_{re} \end{bmatrix} \quad (7)$$

2.3 Direct Torque Control In direct torque control (DTC) ⁽¹⁾⁽¹³⁾, the torque and the magnitude of magnetic flux are kept within the tolerance bounds. With DTC, an instantaneous inverter switching pattern is selected at each instance of sampling according to a predetermined switching table. The configuration of DTC is illustrated in Fig. 3. The flux vector $\psi_{\alpha\beta}$, and the torque and the flux denoted by τ and ψ , respectively, along the static α - β axis are estimated using measured voltage and current obtained using the following equations.

$$\psi_{\alpha\beta} = \int_0^t (v_{\alpha\beta} - Ri_{\alpha\beta})dt + \psi_{\alpha\beta}|_{t=0}, \quad (8)$$

$$\tau = P_n(\psi_{\alpha}i_{\beta} - \psi_{\beta}i_{\alpha}), \quad (9)$$

$$\psi = (\psi_{\alpha}^2 + \psi_{\beta}^2)^{\frac{1}{2}}, \quad (10)$$

where $\psi_{\alpha\beta} = [\psi_{\alpha} \ \psi_{\beta}]'$, $v_{\alpha\beta} = [v_{\alpha} \ v_{\beta}]'$ and $i_{\alpha\beta} = [i_{\alpha} \ i_{\beta}]'$ are the flux, voltage and current in the α - β axis, respectively.

The estimated torque τ and magnetic flux ψ are compared with the reference values τ^{ref} and ψ^{ref} , respectively. Then, the output of each hysteresis comparator is given as follows

$$T = \begin{cases} 0 & \text{if } \tau^{\text{ref}} - \tau < -\varepsilon_{\tau}, \\ 1 & \text{if } \tau^{\text{ref}} - \tau > \varepsilon_{\tau} \text{ and } \end{cases} \quad (11)$$

$$\Psi = \begin{cases} 0 & \text{if } \psi^{\text{ref}} - \psi < -\varepsilon_{\psi}, \\ 1 & \text{if } \psi^{\text{ref}} - \psi > \varepsilon_{\psi}, \end{cases} \quad (12)$$

where ε_{τ} and ε_{ψ} are the tolerance bounds of the torque and magnetic flux, respectively. Then, an input voltage is selected from $\{V_{(000)}, \dots, V_{(111)}\}$ in accordance with the predetermined switching table at each instance of sampling. Unlike vector control, a fast torque response can be achieved by employing the instantaneous inverter switching pattern for DTC. Additionally, the average number of switching operations can be reduced by adjusting the tolerance bounds of the hysteresis comparators.

In DTC for the PMSM, the magnetic flux varies, even when

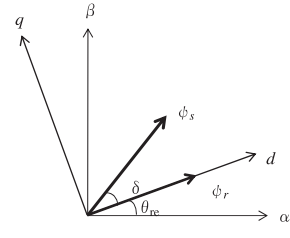
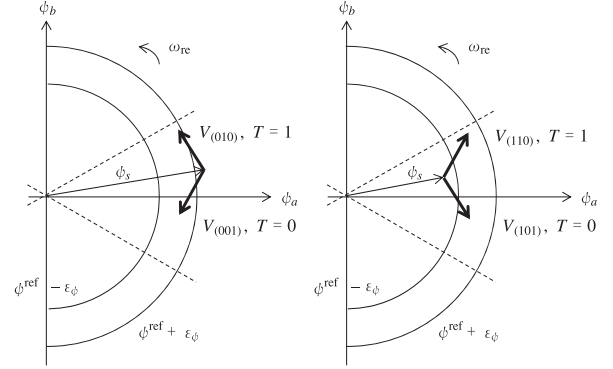

 Fig. 4. Rotor flux ψ_r and stator flux ψ_s in α - β and d - q axis


Fig. 5. Flux vector space for creating switching table

Table 2. Switching table

Ψ	T	θ_s					
		θ_1	θ_2	θ_3	θ_4	θ_5	θ_6
0	0	(001)	(101)	(100)	(110)	(010)	(011)
	1	(010)	(011)	(001)	(101)	(100)	(110)
1	0	(101)	(100)	(110)	(010)	(011)	(001)
	1	(110)	(010)	(011)	(001)	(101)	(100)

the zero voltage vectors are applied, since the magnet rotates with the rotor. Therefore, the zero voltage vectors are usually not used in conventional DTC. Generally, torque is related to the torque angle between the rotor flux vector $\psi_r = K_e$ and the stator flux vector ψ_s , denoted by δ , as shown in Fig. 4. The relationship is given as

$$\tau = \frac{P_n|\psi_s|}{2L_dL_q} [2K_eL_q \sin \delta - |\psi_s|(L_q - L_d) \sin 2\delta]. \quad (13)$$

Torque increases as δ increases ⁽³⁾. That is, the voltage vector is applied in the rotation direction in order to increase the torque, while it is applied in the reverse rotation direction in order to reduce the torque.

The switching table in DTC is composed on the basis of the flux vector space shown in Fig. 5. In Fig. 5, $\psi^{\text{ref}} + \varepsilon_{\psi}$ and $\psi^{\text{ref}} - \varepsilon_{\psi}$ are depicted by the outer and inner circles, respectively. For explanation, let us assume that the rotor rotates counterclockwise and the flux vector position θ_s is in the region θ_1 , as shown in Fig. 2. In the case of the left plot in Fig. 5, either $V_{(001)}$ or $V_{(010)}$ is applied to bring the magnetic flux to the inner circle so that the magnetic flux is decreased. On the contrary, in the case of the right plot in Fig. 5, either $V_{(101)}$ or $V_{(110)}$ is applied to extend the magnetic flux to the outer circle in order to increase the magnetic flux. The other voltage vectors are determined for the other regions θ_2 to θ_6 in a similar manner. The resulting switching table is composed as listed in Table 2.

2.4 Model Predictive Control In model predictive control (MPC) ⁽¹⁵⁾⁽¹⁶⁾, N -step-ahead behavior of controlled

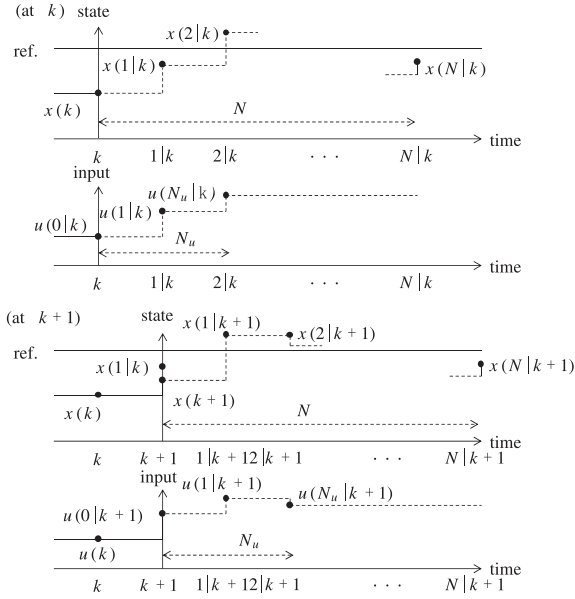


Fig. 6. Illustrative outline of MPC

variables is predicted using the system model. An objective function is usually set so that the predictive values approach the reference. In MPC, various constraints can be also taken into account while minimizing an evaluation of the given objective function.

Figure 6 shows an illustrative outline of the MPC. In Fig. 6, $x(k)$ is the state measured at k , while $x(j|k)$ ($j = 0, \dots, N-1$) denotes the predicted state j steps ahead from k . The positive integer N is called the predictive horizon. The aim is to find input $u(k)$ at k . The optimal input sequence is calculated $\{u^*(0|k), u^*(1|k), \dots, u^*(N_u-1|k)\}$ over $[k, k+N_u]$ (generally, $N_u \leq N$) which minimizes the evaluation of the objective function subject to the constraints. As for the optimal input sequence, only the first entry of $u^*(0|k)$ is applied to the system as $u(k)$ and is held until the next instance of sampling, as shown at the bottom of Fig. 6. The process is iterated at each instance of sampling receding the predictive horizon.

Even if the prediction is not perfectly correct owing to modeling and/or measurement errors, the prediction is recalculated according to the latest measurements at each instance sampling. Therefore, MPC functions well as long as the predicted model is admissible.

3. Model Predictive Direct Torque Control (MPDTC)

In this section, we propose a model predictive direct torque control (MPDTC) approach for the PMSM.

3.1 Configuration of MPDTC The proposed MPDTC system is shown in Fig. 7. In the system, the MPDTC component selects an optimal voltage vector from the finite voltage vector set $\{V_{(000)}, \dots, V_{(111)}\}$ at each instance of sampling.

Let us present a brief outline of MPDTC using Fig. 8. In Fig. 8, suppose that the torque $\tau(k)$ is calculated at k using the measured values. Then, the future evolution of $\tau(j|k)$ ($j = 0, \dots, N-1$)[†] is predicted according to Eqs. (2) and (3). Recall that $v_{dq}(j|k) \in C_{uw}^{dq}(\theta_{re}(j|k))\{V_{(000)}, \dots, V_{(111)}\}$. That

[†] Hereafter, the notation of ($j = 0, \dots, N-1$) is skipped unless misunderstanding.

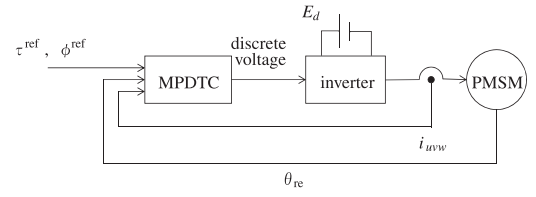


Fig. 7. Configuration of MPDTC

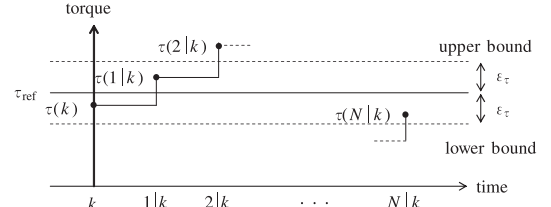


Fig. 8. Outline of MPDTC

is, one of the eight discrete voltage vectors is selected at each instance of sampling. Similarly, the flux $\psi(j|k)$ is predicted according to Eqs. (2) and (4).

In general, frequent switching leads to fast torque and flux responses. However, frequent switching results in loss in the devices of the inverter. Therefore, we also take a number of switching operations into account apart from torque and flux. Define the objective function in MPDTC as

$$J(i_{dq}(k)) = \sum_{j=0}^{N-1} \|s(j|k) - s(j-1|k)\|^2 + Q_{\tau} p_{\tau}(j|k) + Q_{\psi} p_{\psi}(j|k), \dots \quad (14)$$

$$p_{\tau}(j|k) = \begin{cases} 0 & \text{if } |\tau^{ref} - \tau(j|k)| \leq \epsilon_{\tau}, \\ \left(\frac{\tau^{ref} - \tau(j|k)}{\epsilon_{\tau}}\right)^2 & \text{if } |\tau^{ref} - \tau(j|k)| > \epsilon_{\tau}, \end{cases} \dots \quad (15)$$

$$p_{\psi}(j|k) = \begin{cases} 0 & \text{if } |\psi^{ref} - \psi(j|k)| \leq \epsilon_{\psi}, \\ \left(\frac{\psi^{ref} - \psi(j|k)}{\epsilon_{\psi}}\right)^2 & \text{if } |\psi^{ref} - \psi(j|k)| > \epsilon_{\psi}, \end{cases} \dots \quad (16)$$

where Q_{τ} and Q_{ψ} are the appropriate weights. In Eq. (14), the first term is the number of switching operations over $[k, k+N]$, and the second and third terms are the penalties when the predictive torque and/or flux are not within the given tolerance bounds. Then, the optimization problem is formalized as

$$\{s^*(0|k), \dots, s^*(N-1|k)\} = \underset{\{s(0|k), \dots, s(N-1|k)\}}{\operatorname{argmin}} J(i_{dq}(k)) \dots \quad (17)$$

subject to

$$\begin{cases} s(-1|k) = s(k-1), & i_{dq}(0|k) = i_{dq}(k), \\ \theta_{re}(j|k) = \theta_{re}(k) + j\omega_{re}T_s, \\ i_{dq}(j+1|k) = A i_{dq}(j|k) + B v_{dq}(j|k) + f(k), \\ \tau(j|k) = P_n [K_e i_q(j|k) + (L_d - L_q) i_d(j|k) i_q(j|k)], \\ \psi(j|k) = [(K_e + L_d i_d(j|k))^2 + (L_q i_q(j|k))^2]^{\frac{1}{2}}, \\ v_{dq}(j|k) = C_{uw}^{dq}(\theta_{re}(j|k)) \frac{E_d}{2} s(j|k), \\ s(j|k) \in \{-1, 1\}^3, \\ (\forall k, j) \end{cases} \dots \quad (18)$$

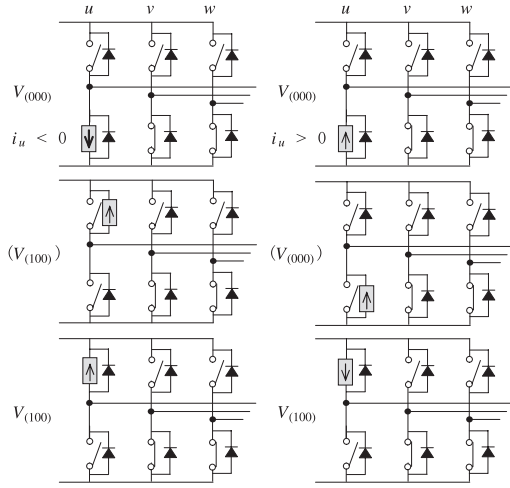


Fig. 9. Dead-time of inverter (top: before switching, middle: dead-time (if any), and bottom: after switching)

Note that, $\frac{E_d}{2} s(jk) \in \{V_{(000)}, \dots, V_{(111)}\}$ in Eq. (18) is always feasible, as it is one of the discrete voltage vectors which are consistently generated in the inverter.

For obtaining an optimal solution $\{s^*(0|k), s^*(1|k), \dots, s^*(N-1|k)\}$ in Eq. (17), only the first entry $s^*(0|k)$ is adopted as $v_{dq}(k) = C_{uvw}^{dq}(\theta_{re}(k)) \frac{E_d}{2} s^*(0|k)$ at k and held until $k+1$. The optimization is iterated at $k+1$ using the available measurements at $k+1$ receding the predictive horizon N . MPDTC approach enables us to achieve both faster torque and flux responses with fewer switching operations in the inverter while satisfying the imposed constraints.

3.2 Treatment of Dead-time in Inverter Model and Rotation of Rotor During a sampling period, the d - q voltage is no longer constant since it depends on the electrical angle, θ_{re} . Thus, the voltage due to the dead-time and the electrical angle needs to be taken into account to ensure prediction accuracy in MPDTC.

In practice, the dead-time is required to avoid shorting in the inverter regardless of use of DTC or MPDTC. The output voltage during the dead-time, which depends on the direction of the phase current, may lead to prediction errors. Thus, it is necessary to incorporate the conditions during the dead-time into the predictive model in order to improve the control performance of MPDTC.

The dead-time in the inverter is explained using Fig. 9. Let us consider the case where the switching pattern $V_{(000)}$ is switched to $V_{(100)}$, which requires us to turn on the upper arm in the u -phase. If the u -phase current is negative before switching, as shown in the top left in Fig. 9, current flows through the diode at the instant of switching, as shown in the middle left part of Fig. 9. The situation is equivalent to $V_{(100)}$. Then, the switching pattern can be instantaneously changed at the switching. On the other hand, if the u -phase current is positive before switching, the current during the dead-time is as shown in the middle right part of Fig. 9, which is equivalent to $V_{(000)}$. Thus, the voltage vector during the dead-time is considered. As described above, the voltage vector during the dead-time is determined according to the signs of the phase current. In this example, when the switching from $V_{(000)}$ to $V_{(100)}$ is required, the dead-time voltage vector is $V_{(000)}$, only when the u -phase current is positive at the

Table 3. Dead-time voltage vector and sign of phase current (from $v_{uvw}(j-1|k) = V_{(000)}$ to $v_{uvw}(j|k) = V_{(100)}$)

$\text{sgn}(i_u(jk))$	$\text{sgn}(i_v(jk))$	$\text{sgn}(i_w(jk))$	$v_{uvw}^{\text{DT}}(jk)$
+	+	+	$V_{(000)}$
+	+	-	$V_{(000)}$
+	-	+	$V_{(000)}$
+	-	-	$V_{(000)}$
-	+	+	$V_{(100)}$
-	+	-	$V_{(100)}$
-	-	+	$V_{(100)}$
-	-	-	$V_{(100)}$

instance of switching. For the other two phases, the equivalent dead-time voltage vectors are listed in Table 3. In the same way, the dead-time voltage vector is determined for the other switching patterns according to the signs of the associated phase current.

Next, let us evaluate the voltage vector when the magnetic pole position is rotating. Each vector in $\{V_{(000)}, \dots, V_{(111)}\}$ rotates synchronously with the rotation of the motor in the d - q axis, while it is fixed in the u - v - w axis. Thus, the predicted voltage vector j steps ahead from k is not fixed until the next step. The difference also leads to a prediction error, which appears especially when driven at relatively close speed to the base speed. To solve the problem, the method detecting the magnetic pole in PWM current control has been proposed⁽²⁹⁾. In MPDTC, the method is adopted for improving the prediction accuracy.

For the purpose, let us denote a candidate voltage vector j steps ahead from k using $v_{uvw}(j|k) \in \{V_{(000)}, \dots, V_{(111)}\}$. The dead-time voltage corresponding to $v_{uvw}(j|k)$, is denoted by $v_{uvw}^{\text{DT}}(j|k)$. Note that $v_{uvw}^{\text{DT}}(j|k)$ is uniquely defined according to the signs of $v_{uvw}(j|k)$. If the dead-time is required at $k+j$, the interval $[(k+j)T_s, (k+j+1)T_s]$ is partitioned into $[(k+j)T_s, (k+j)T_d]$ and $[(k+j)T_d, (k+j+1)T_s]$ to which $v_{uvw}^{\text{DT}}(j|k)$ and $v_{uvw}(j|k)$ are assigned, respectively, where T_d is the dead-time. The interval $[(k+j)T_s, (k+j+1)T_s]$ is not partitioned unless the dead-time is required.

Now, we derive $\bar{v}_{dq}(j|k)$ for prediction in MPDTC by averaging the voltage trajectory during one sampling period as follows.

$$\begin{aligned}
 \bar{v}_{dq}(j|k) &= \frac{1}{T_s} \left[\int_0^{T_d} C(\theta_{re}(j|k) + \omega_{re}\tau) v_{uvw}^{\text{DT}}(j|k) d\tau \right. \\
 &\quad \left. + \int_{T_d}^{T_s} C(\theta_{re}(j|k) + \omega_{re}\tau) v_{uvw}(j|k) d\tau \right] \\
 &\approx \frac{T_d}{T_s} C_{uvw}^{dq} \left(\theta_{re}(j|k) + \frac{\omega_{re}T_d}{2} \right) v_{uvw}^{\text{DT}}(j|k) \\
 &\quad + \frac{T_s - T_d}{T_s} C_{uvw}^{dq} \left(\theta_{re}(j|k) + \frac{\omega_{re}(T_s + T_d)}{2} \right) v_{uvw}(j|k). \\
 &\dots\dots\dots (19)
 \end{aligned}$$

In Eq.(19), the last equation is derived since T_s and T_d are sufficiently small. In addition, in Eq.(19), $v_{uvw}(j|k)$, $v_{uvw}^{\text{DT}}(j|k) \in \{V_{(000)}, \dots, V_{(111)}\}$. Then, when the dead-time is taken into account, $v_{dq}(j|k)$ in Eq. (18) is replaced by the following two equations.

$$v_{dq}(j|k) = \frac{T_d}{T_s} C_{uvw}^{dq} \left(\theta_{re}(j|k) + \frac{\omega_{re} T_d}{2} \right) \frac{E_d}{2} s^{DT}(j|k) + \frac{T_s - T_d}{T_s} C_{uvw}^{dq} \left(\theta_{re}(j|k) + \frac{\omega_{re}(T_s + T_d)}{2} \right) \frac{E_d}{2} s(j|k), \dots \dots \dots (20)$$

$$s^{DT}(j|k) = D(\text{sgn}(i_u(j|k), i_v(j|k), i_w(j|k)), s(j-1|k), s(j|k)), \dots \dots \dots (21)$$

where D is the map used to provide a unique switching pattern for the dead-time as listed in Table 3.

4. Simulations and Experiments

In this section, the proposed MPDTC approach is compared with conventional DTC. For convenience, the proposed basic approach is denoted by MPDTC(I), and the other approach is denoted by MPDTC(II) which treats with the dead-time voltage vector and the rotation of the motor as introduced in Sect.3.2. The effectiveness of the proposed approaches is verified through simulations and experiments.

4.1 Simulations In the simulations, MPDTC(I) is compared with the conventional DTC for clearly examining the characteristics of the former. All simulations are performed using in MATLAB ver. 7.0.1 (R14) SP1 (Math Works Inc.). The parameters are listed in Table 4. The dead-time is not taken into account for the simulations.

In MPDTC(I), the weights in Eq. (14) are $Q_\tau = Q_\psi = 4$, respectively. The weights are set to balance the first term in Eq. (14) related to the number of switching operations at each instance of sampling, which takes at most three if all switching devices in the inverter are switched. That is, the number of switching operations is weighted as long as both the torque and the flux are within the tolerance bounds. On the other hand, if either the torque or flux is beyond its tolerance bounds, a trade-off is achieved between the number of switching operations and the torque or flux in order to bring them within the tolerance bounds.

We investigate the step responses of the torque and the flux at a constant speed for DTC and MPDTC(I). The torque and flux tolerances in DTC are set to be smaller than those in MPDTC(I) in order to make the comparison conditions equivalent.

Figures 10 and 11 show the simulation results of the torque

and the flux responses, respectively. From Figs. 10 and 11, it can be observed that the torque and the flux responses by MPDTC(I) are better than those by DTC.

Figure 12 shows the simulation result of the selected switching vectors. In steady-state, MPDTC(I) selects $V_{(000)}$ and $V_{(111)}$, which are zero vectors. On the other hand, their voltage vectors are not selected since the two-level hysteresis comparator is used in DTC. Therefore, MPDTC(I) achieves faster torque and flux responses and a smaller number of switching operations than DTC.

4.2 Experiment

In experiments, we first compare MPDTC(I) with DTC. Then, MPDTC(II) is compared with MPDTC(I).

4.2.1 DTC and MPDTC(I)

The configuration of the experimental system is shown in Fig. 13. The numerical operations are executed in PE-Expert3 embedded with DSP (Myway Plus Corp.). The dead-time is negligibly small in the experiments as it is 20 [ns] while the sampling period is $T_s = 100 [\mu s]$, which aims comparison of MPDTC(I) with DTC. The parameters in the experiments are the same as those in the simulations.

Figures 14 and 15 show the experimental results of the torque and the flux responses, respectively. In Fig. 14, the torque reference τ^{ref} is changed from 1 [Nm] to 5 [Nm] at 10 [ms] at the speed of 300 [rpm]. The flux reference ψ^{ref} is also changed, as shown in Fig. 15, with τ^{ref} on the basis of the maximum torque per ampere (MTPA)⁽³⁰⁾. From Fig. 14, the torque ripple with MPDTC(I) is smaller than that with DTC.

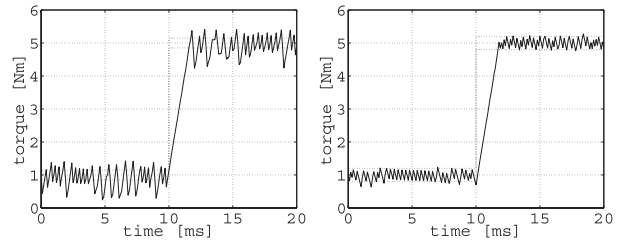


Fig. 10. Simulation of estimated torque response (left: DTC, right: MPDTC(I))

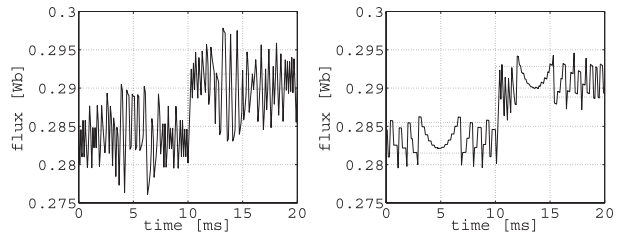


Fig. 11. Simulation of estimated magnetic flux (left: DTC, right: MPDTC(I))

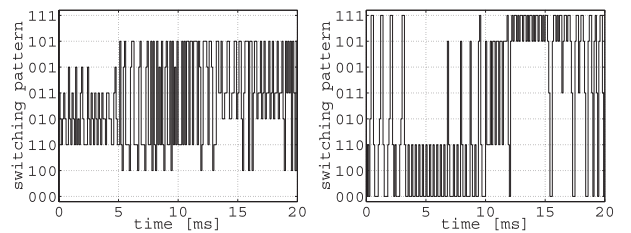


Fig. 12. Simulation of switching vectors (left: DTC, right: MPDTC(I))

Table 4. Parameters

rated power	1.5 [kW]
dc-link voltage E_d	100 [V]
stator resistance R_d	0.681 [Ω]
inductance (d -axis) L_d	10.0 [mH]
inductance (q -axis) L_q	15.2 [mH]
e.m.f. constant K_e	0.2832 [Wb]
base speed	850 [rpm]
pairs of rotor poles P_n	3
torque tolerance (DTC) ε_τ	0.15 [Nm]
flux tolerance (DTC) ε_ψ	0.001 [Wb]
torque tolerance (MPDTC(I)) ε_τ	0.2 [Nm]
flux tolerance (MPDTC(I)) ε_ψ	0.002 [Wb]
sampling period T_s	100 [μs]
prediction horizon N	1

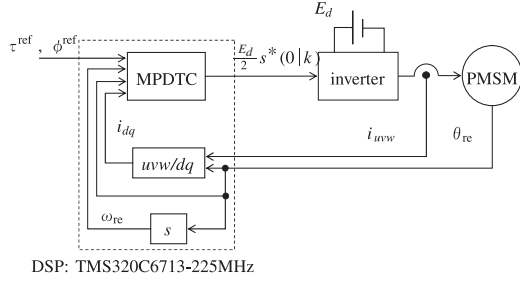


Fig. 13. Experimental system

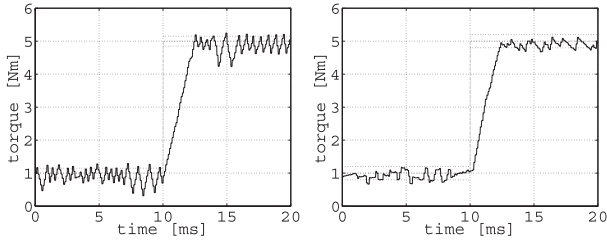


Fig. 14. Experimental results for estimated torque response (without dead-time) (left: DTC, right: MPDTC(I))

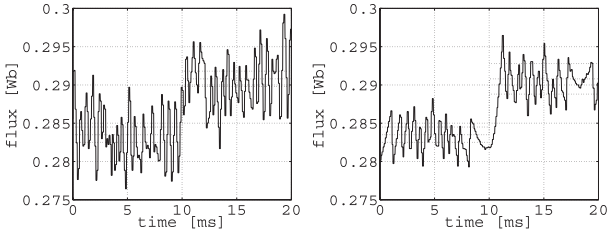


Fig. 15. Experimental results for estimated magnetic flux (without dead-time) (left: DTC, right: MPDTC(I))

The torque response of MPDTC(I) is more effective than that of DTC which resorts to the fixed switching table since, in MPDTC(I), the switching pattern in the inverter is selected according to the predicted values online. MPDTC(I) is also superior to DTC in the flux response as well, as shown in Fig. 15.

For an available torque response $\tau(k)$ and the tolerance ε_τ , define

$$p_\tau(k) = \begin{cases} 0 & \text{if } |\tau_{ref} - \tau(k)| \leq \varepsilon_\tau, \\ \left(\frac{\tau_{ref} - \tau(k)}{\varepsilon_\tau} \right)^2 & \text{if } |\tau_{ref} - \tau(k)| > \varepsilon_\tau. \end{cases} \quad (22)$$

Then, the torque variance is calculated as $\sum_k p_\tau(k)$. In a similar way, τ is replaced by ψ in Eq. (22) for the flux variance.

Figure 16 shows the quantitative evaluation of the performance of MPDTC(I) and DTC at different operating points, in addition to the MPDTC(I) to DTC ratios for the average number of switching operations, torque variance, and flux variance. Each ratio is a value with MPDTC(I) to that with DTC. The selected operating points are all the combinations of torque and speed values in $\{0, 1, \dots, 5\}$ and $\{100, 200, \dots, 500\}$, respectively. From Fig. 16, it can be seen that all the ratios are less than one, which implies that MPDTC(I) is superior to DTC. The average number of switching operations performed with MPDTC(I) at almost all operating points is less than that performed with DTC, except

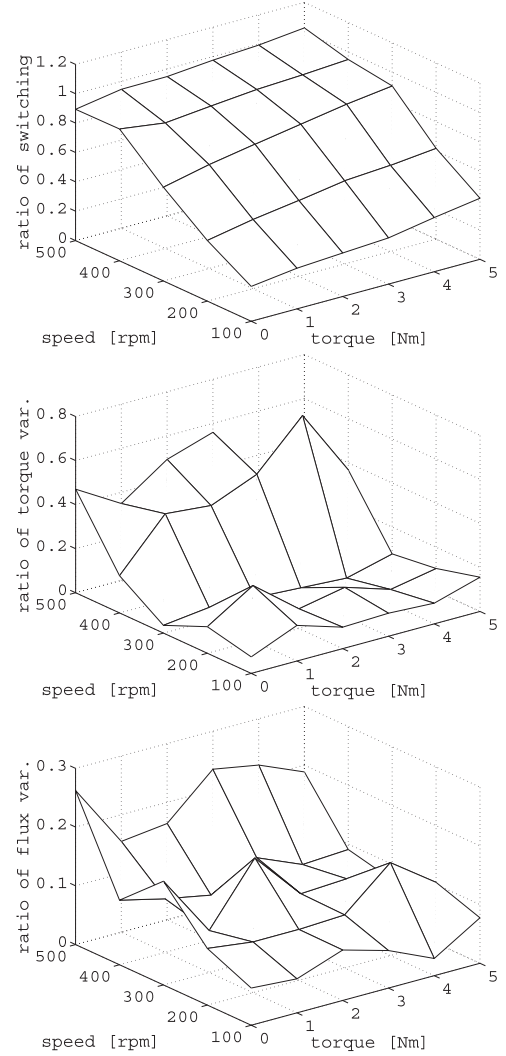


Fig. 16. Evaluation of DTC and MPDTC(I) (top: ratio of average number of switching operations, middle: ratio of torque variance, bottom: ratio of flux variance)

Table 5. Evaluation of DTC and MPDTC(I) at selected operating points (without dead-time) (average)

	DTC	MPDTC(I)
number of switching operations [s^{-1}]	14025	9103
torque variance [$N^2 m^2$]	1.04×10^{-2}	0.75×10^{-2}
flux variance [Wb^2]	4.92×10^{-6}	5.32×10^{-7}

for the points at the rotational speed of 500 [rpm] and the torque reference of 5 [Nm]. As shown in the middle and bottom plots in Fig. 16, the torque and the flux variances with MPDTC(I) are effectively reduced when compared to those with DTC.

Table 5 lists the average evaluation at selected operation points. From Table 5, the average number of switching operations with MPDTC(I) is less than that with DTC, and the torque and the flux variances are also lower for MPDTC(I).

In addition, it can be also observed that the current ripple by MPDTC(I) is smaller than that by DTC even though the d - q current is not evaluated directly in both the methods, which is not shown as it is not the main content in this paper. As a result, the ripple of the estimated torque behaves alike.

4.2.2 MPDTC(I) and MPDTC(II) We demonstrate

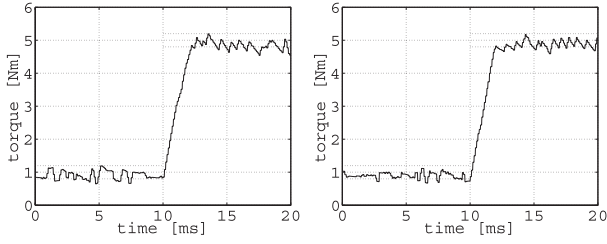


Fig. 17. Experimental results for estimated torque response (with dead-time) (left: MPDTC(I), right: MPDTC(II))

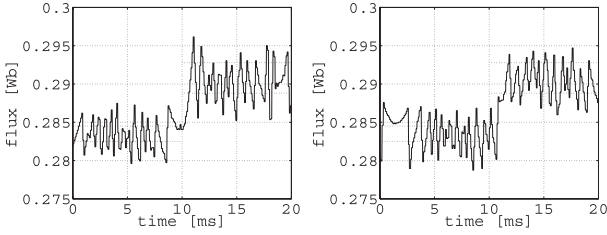


Fig. 18. Experimental results for estimated magnetic flux (with dead-time) (left: MPDTC(I), right: MPDTC(II))

Table 6. Evaluation of MPDTC(I) and MPDTC(II) at selected operating points (with dead-time) (average)

	MPDTC(I)	MPDTC(II)
number of switching operations [s^{-1}]	9331	9443
torque variance [N^2m^2]	1.04×10^{-2}	0.75×10^{-2}
flux variance [Wb^2]	7.04×10^{-7}	5.09×10^{-7}
torque offset [Nm]	2.64×10^{-2}	1.91×10^{-2}
flux offset [Wb]	1.79×10^{-7}	1.36×10^{-7}

the accuracy of the predictive model in MPDTC described in the previous section. The dead-time is set as $4.0 \mu s$. Note that MPDTC(II) takes the dead-time vector and the rotation of the rotor into account, which are not considered explicitly by MPDTC(I).

Figures 17 and 18 show the experimental results of the torque and the flux responses at 300 [rpm], respectively. There exists little difference between MPDTC(I) and (II) from Figs. 17 and 18.

To examine the results for other operating points, the ratios of the average number of switching operations, and the torque and flux variances for some of the operating points are shown in Fig. 19. The selected operating points are the same as in the previous subsection. From Fig. 19, it can be observed that with the exception of zero torque conditions, the ratio of the average number of switching operations is nearly one. This is because the switching pattern by MPDTC(II) is determined on the basis of the signs of the phase current around the reference of 0 [Nm], namely 0 [A]. The ratios of torque and flux variances are smaller than one in almost all operating points.

Table 6 lists the evaluation in selected operating points. Note that the values of MPDTC(I) listed in Table 5 are not equal to those listed in Table 6 since the former and the latter are obtained under the conditions with the dead-time and without it, respectively. From Table 6, the average number of switching operations for MPDTC(I) and MPDTC(II) is almost equal. On the other hand, the variances of the torque and the flux as well as their offsets with MPDTC(II) are less than those with MPDTC(I). Thus, MPDTC(II) is superior to

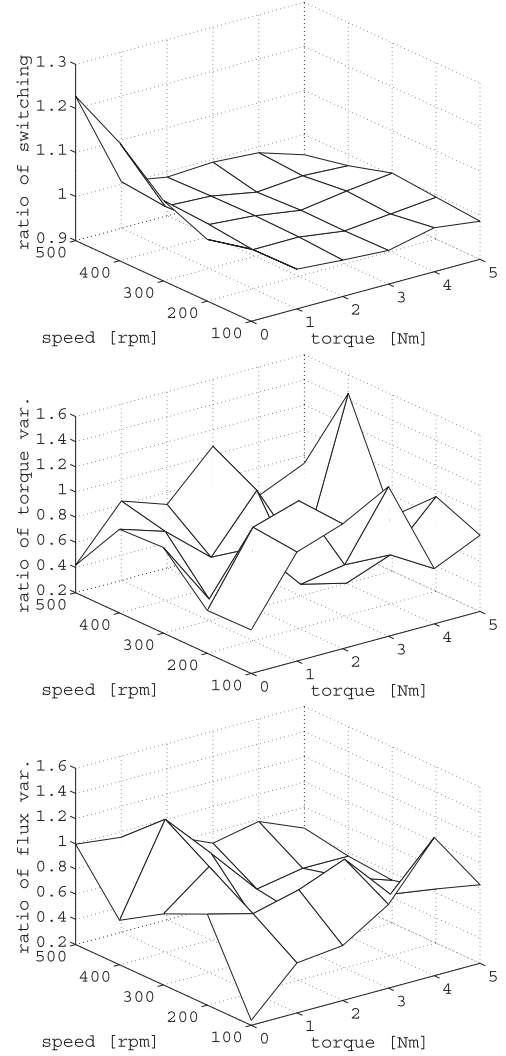


Fig. 19. Evaluation of MPDTC(I) and MPDTC(II) (top: ratio of average number of switching operations, middle: ratio of torque variance, bottom: ratio of flux variance)

MPDTC(I) since the dead-time vector and the rotation of the rotor are explicitly considered in MPDTC(II).

5. Conclusion

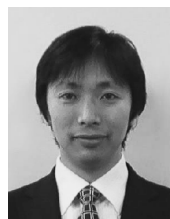
In this paper, we proposed a model predictive direct torque control (MPDTC) approach for permanent magnet synchronous motors. In the proposed approach, the number of switching operations was considered together with torque and flux responses. In addition, the behavior during the dead-time and the rotation of the rotor were also addressed in our approach to ensure accurate prediction. The effectiveness of the proposed method was verified through simulations and experiments. Both the simulations and experiments established that the torque and the flux ripple in steady-state decreased with a smaller number of switching operations in MPDTC when compared with conventional DTC at selected operating points, thus achieving fast transient responses for stepwise changes in the reference values. We also showed that with the exception of the lower torque condition, the variances and offsets in the torque and flux responses were reduced when the dead-time and the rotation of the rotor were taken into account in the prediction model.

Future subjects include implementation of the proposed approach for a longer prediction horizon, which requires the use of an efficient algorithm for optimization calculation. Further experimental verification is needed for a smaller sampling time and a relatively large dead-time in order to demonstrate MPDTC(II) which considers the dead-time vector since it could not be implemented due to the calculation in our experimental system. In addition, the offset in the torque response is dependent on the speed of the motor. To reduce the offset, we are presently developing an appropriate setting for the reference values.

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